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To cite this article: Yaofeng Xue, Kun Wang & Yisheng Qiu (2025) Enhancing Online Learning: A Multimodal Approach for Cognitive Load Assessment, International Journal of Human-Computer Interaction, 41:4, 2692-2702, DOI: [10.1080/10447318.2024.2327198](https://doi.org/10.1080/10447318.2024.2327198)

To link to this article: <https://doi.org/10.1080/10447318.2024.2327198>



Published online: 22 Mar 2024.



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Enhancing Online Learning: A Multimodal Approach for Cognitive Load Assessment

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ABSTRACT

Online learning has become increasingly popular in recent years, but the frequent occurrence of cognitive overload has been notably impacting both the learning experience and effectiveness. Therefore, based on optimizing online learning, this study proposes a research framework for cognitive load assessment of online learning based on three modal data: electroencephalography (EEG), eye tracking, and face. Following this framework, a neural network was used to construct a cognitive load assessment model for online learning that integrates multimodal data. After validation, the assessment accuracy of the model reaches 91.52%. In addition, the results based on multimodal data analysis can be used as a reference for the development of learning resources and the optimization of online courses in intelligent online learning platforms. The assessment model constructed in this study can also be applied to the online learning platform, which is expected to realize prescription-adaptive online learning based on cognitive load assessment. Due to current research limitations, only specific thematic learning models have been explored. Future research will focus on model fine-tuning, complex learning scenarios and themes designing and expansion of research scale to enhance the model's generalization capabilities.

KEYWORDS

Multimodal data; online learning; cognitive load; eye tracking; EEG signals; facial action recognition

1. Introduction

Cognitive load refers to the amount of work that the cognitive system exerts on working memory when humans perform a particular task (Sweller, 1988; Paas & Van Merriënboe, 1994), and it is affected by a variety of factors, including task difficulty, skill level, age, and so on (Paas et al., 2003). Cognitive load results from the limited capacity and duration of working memory during the performance of a task. This may lead to under- or overloading of cognitive load, both of which are detrimental to task performance, especially in the case of overloading (Hussain et al., 2013).

Since the beginning of the 21st century, with the rapid development of various scientific and technological advancements, online learning has gradually taken shape and become widespread. The COVID-19 pandemic has further accelerated this trend, significantly increasing the reliance on and the prevalence of online learning platforms worldwide. This mode of learning not only breaks the constraints of time and space but also provides learners with a wealth of learning resources. However, it also comes with certain drawbacks. The abundance of information in online learning and its high demands on learners' capabilities often lead to phenomena such as cognitive overload and poor learning engagement (Chen et al., 2005; Warrick, 2021). Existing research has shown that maintaining cognitive load within

an appropriate range can motivate learners and enhance their learning engagement, thereby helping them achieve better learning outcomes and experiences (Kirschner et al., 2018).

To achieve this goal, accurately and conveniently assessing the cognitive load of online learners becomes an indispensable research task. While research on assessing cognitive load has been ongoing since the last century, commonly used methods involve post-assessment through various subjective scales or questionnaires, requiring learners to recall their learning processes and subjectively evaluate their cognitive load.

However, it is undeniable that such assessment methods are highly subjective and susceptible to bias, making them unsuitable for the online learning environment. In the context of online learning, learners complete their learning independently, and teachers are unable to grasp the learner's state and cognitive load information well. The learning system often presents learning resources rigidly and cannot adaptively provide suitable resources for students. In order to achieve adaptive resource recommendation, objective multimodal data such as eye movements, brainwaves, and facial expressions can be used. When learners sit in front of a computer to learn, this data is easily accessible. Multimodal data can complement each other's shortcomings, and data fusion can improve the accuracy of the

model. Multimodal analysis methods have been widely used in various aspects such as daily behavioral activities (Luo et al., 2022), driver monitoring (Zhang, Feng, et al., 2022; Zhang, Guo, et al., 2022), and patient diagnosis (Goodwin & Harabagiu, 2016). Meanwhile, multimodal research in online learning is gradually gaining attention.

The research questions guiding this study were as follows:

1. What is the cognitive load evaluation model in online learning?
2. How effective is the model constructed using multimodal data?

Therefore, this study discovered and proposed an online learning cognitive load model based on multimodal data. This model relies on three modalities of data generated by learners during the learning process: eye-tracking, facial, and electroencephalogram data, enabling a more reliable and real-time assessment of cognitive load.

2. Related work

2.1. Traditional cognitive load assessment

Research on cognitive load originated in the last century, and initially, methods for assessing cognitive load primarily relied on scales. The application of these methods was based on the premise that individuals can rate their cognitive resource consumption during cognitive processes, and this resource consumption is related to factors such as task difficulty, mental effort, and time pressure. Commonly used scales include the Cooper-Harper Scale (CH) (Cooper, 1990), the Workload Profile Index Ratings (WP) scale (Chen et al., 2005), and the National Aeronautics and Space Administration-Task Load Index (NASA-TLX) scale (Hart, 2006). Among these, the CH scale involves participants directly rating their overall cognitive load, with higher scores indicating higher cognitive load. In contrast, the WP and NASA-TLX scales indirectly assess cognitive load from multiple dimensions, calculating the magnitude of cognitive load through the weighting of different dimensions. In the early development of the cognitive load research field, these methods were widely used. However, with the advancement of modern science and technology, these methods also exhibited limitations. These methods all fall under the category of post-assessment subjective evaluation methods. In educational settings, learners are required to recall their learning processes and self-report cognitive load using scales after completing learning tasks. Such assessment methods are clearly too subjective and susceptible to bias, and they cannot provide real-time evaluation, making them unsuitable for online learning environments. Therefore, this study opts for a comprehensive measurement approach that combines scales with sensor technology. This integrated measurement approach for assessing cognitive load can effectively enhance the accuracy and validity of measurements.

2.2. Data-based cognitive load assessment

Compared to subjective scales, comprehensive measurement methods are more objective. In the field of cognitive load assessment research, objective data-driven measurements are mostly concentrated in the human-computer interaction domain. However, there is a scarcity of cognitive load assessments tailored to specific learning contexts in the education field. Eye-tracking technology is one common method, which assesses cognitive load by collecting eye movement data such as fixation, saccade, and pupil during learning or task processes. Andrzejewska and Skawińska (2020) used eye-tracking technology to measure cognitive load in students during programming learning and found that longer fixation times and shorter saccade amplitude were closely related to cognitive load. Physiological information such as electroencephalography (EEG) (Örüm & Akbulut, 2019), functional near-infrared spectroscopy (fNIRS) (Uysal, 2015), heart rate and pulse (Wang & Guo, 2019) are also gradually being applied to assess cognitive load. Since cognitive load is closely related to the brain, EEG signals exhibit high synchronicity with cognitive load (Tasmi & Mohammad Zavid, 2021), making EEG signal assessment a common approach. Mazher et al. (2017) demonstrated that cognitive load during the learning process can be assessed using EEG signals through feature extraction and partial directed coherence methods.

In recent years, with the maturity, widespread availability, and reduced cost of various sensor devices and technologies, more and more research has attempted to use multimodal data to assess cognitive load. Compared to single-modal data, assessment methods based on the fusion of multimodal data offer higher robustness and noise resistance (Debie et al., 2021). By combining two modalities of perception to predict learner performance, this data can provide appropriate adaptive assistance to enhance the learning experience and optimize their performance (Khedher et al., 2019). By constructing VR simulation learning scenarios and comprehensively utilizing multimodal information collected from the environment, the classification of learner states has been achieved (Bounhar et al., 2022), paving the way for multimodal learning analysis in VR scenarios.

Overall, multimodal fusion can expand dataset size, enrich feature information dimensions, and improve model performance. Therefore, in this study, a research framework for online learning cognitive load assessment based on multimodal data, including EEG, eye-tracking, and facial expression data, was chosen. This research framework was applied to assess the cognitive load levels of learners in online learning contexts.

3. Research framework

This study has developed an online learning cognitive load assessment framework based on multimodal feature fusion. This framework consists of three main components: data collection, feature extraction, and data fusion, as illustrated in Figure 1.

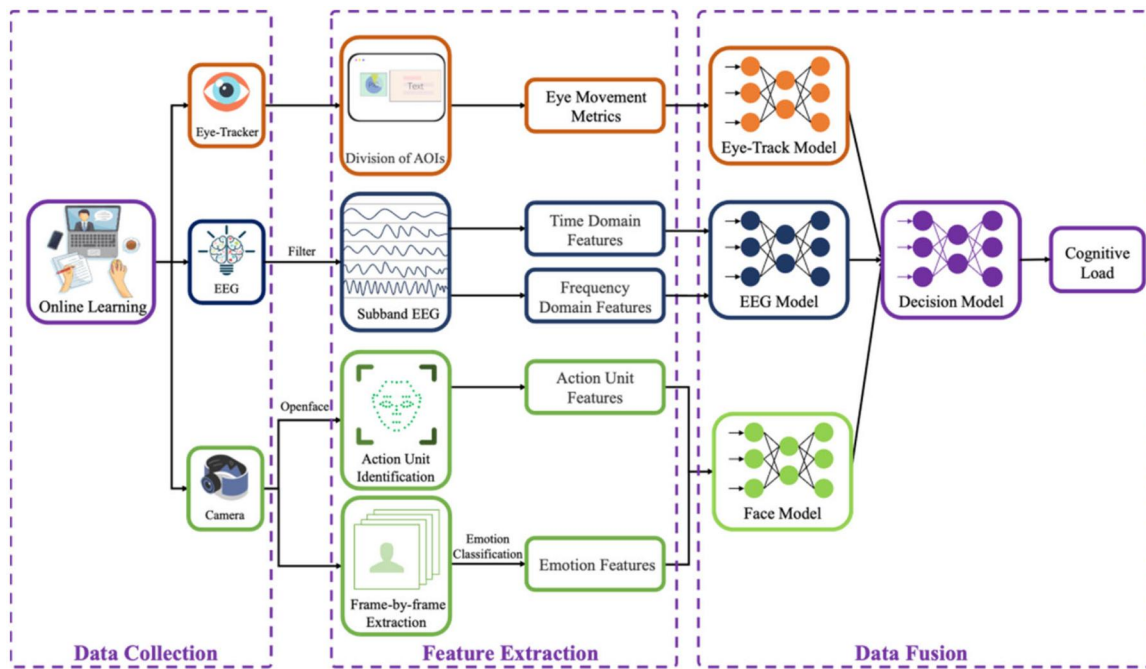


Figure 1. Research framework for online learning cognitive load assessment based on multi-modal data.

3.1. Data collection

Online learning has become widely popular today, and it has evolved into two main forms: synchronous live classes and asynchronous learning based on learning resources (Qiao et al., 2021). The framework developed in this study is specifically designed for the latter, the online learning environment. During the learning process, the primary data collected includes eye-tracking data, EEG signals, and facial data. Firstly, eye-tracking devices are utilized as display devices for presenting learning resources, simultaneously capturing learners' eye-tracking data. The eyes, being directly connected to the brain, implicitly provide crucial information about human cognition (Souchet et al., 2021). Eye-tracking devices are non-invasive and do not interfere with learning behaviors, making them widely applicable in research related to human cognition (Skaramagkas et al., 2023; Tong & Nie, 2022). Secondly, a convenient EEG headset is used to capture changes in frontal lobe EEG signals from learners. The brain, as the direct site of cognitive activities, exhibits changes in EEG signals closely related to cognition (Dong et al., 2016; Mazher et al., 2017). Additionally, a camera is employed to record learners' facial movements in real-time and analyze emotional changes. Analyzing emotions can help study the relationship between emotions, learning motivation, learning styles, cognition, and the impact of emotions on achievement and performance (Yadegaridehkordi et al., 2019).

3.2. Feature extraction

After collecting multimodal data from the relevant devices, it is essential to extract valuable information from the raw data.

3.2.1. Eye-tracking

In eye-tracking research, the presented content is typically divided into different regions known as areas of interest (AOI), based on the arrangement of various visual elements in the stimulus material. Eye-tracking behavioral metrics within these AOIs are then analyzed to understand cognitive states or attention distribution. In the framework proposed in this study, the visual elements within the learning resources can be divided into different AOI, such as text and images. Metrics like fixation count, fixation duration, first fixation time, average pupil diameter, and pupil diameter standard deviation can be calculated for each AOI. Liu, Zhan et al. used the proportional time on the matrix area, rate of toggling, the rate of latency to first toggle instead of absolute values (2023). This is because in the process of eye tracking collection. The collection rate of various indicators varies from person to person, and the calculation of relative ratios is more meaningful. Therefore, we followed this approach in our research.

3.2.2. EEG signals

EEG signals represent the summation of the electrical reactions of numerous neurons in response to brain activity. Electroencephalography (EEG) is a method that involves placing several electrodes on the scalp to collect and record changes in EEG signals during brain activity. The raw EEG signals collected via EEG devices can be filtered to decompose them into five different frequency bands, each of which corresponds to different human cognitive activities (Sulaiman et al., 2022). These frequency bands are Delta (0.5–4 Hz), Theta (4–7 Hz), Alpha (8–12 Hz), Beta (13–30 Hz), and Gamma (>30 Hz) (Zhang et al., 2021). Delta waves have slower frequencies and are more active during deep sleep. Theta waves often occur during

dreaming, fatigue, deep relaxation, or meditation. Alpha waves are typically present when a person is awake and relaxed. Beta waves are more active when a person is focused and attentive, and Gamma waves, with their faster frequencies, appear during advanced cognitive activities. In the framework proposed in this study, features can be extracted from the filtered EEG signals for each frequency band. These features may include mean, standard deviation, maximum, minimum values in the time domain, as well as relative power spectral density, absolute power spectral density, and other features in the frequency domain.

3.2.3. Facial movements and expressions

Facial movements and expressions contain rich information related to learning and cognition. Research has shown that detecting facial movements and expressions can help identify aspects of learners' engagement and attention (Maqableh et al., 2023; Sathik & Jonathan, 2013). In the framework proposed in this study, the facial action detection tool, OpenFace will be used to recognize changes in various action units from the video recorded by the camera (Baltrušaitis et al., 2016). Features such as the mean and standard deviation of the intensity of each action unit can be extracted. Additionally, facial images from each frame of the video will be directly input into a facial expression recognition model (Barsoum et al., 2016), providing a continuous record of learners' expressions. The occurrence frequency of different expression categories can then be counted.

3.3. Data fusion

The scenario of online learning emphasizes more on the use of tools and platforms by learners, which enhances their participation in the learning process and maintains the correlation between learning behavior data and teaching content. This is the research foundation of multimodal data fusion in online learning. Compared to the limited data sources and potential one-sided risks of single modality, multimodality can provide richer information, and multiple modalities can complement each other to form a shared information space, thereby helping to form a more comprehensive understanding and explanation of learning phenomena. (Lahat et al., 2015) There are three common strategies for multimodal data fusion: data-level fusion, feature-level fusion, and decision-level fusion. In this study, the decision-level fusion strategy is chosen. Under this fusion strategy, a sub-assessment model is constructed for each modality of data, with each sub-model being an artificial neural network. The eye-tracking model consists of 3 hidden layers with 32, 16, and 8 neurons, respectively, using the "Relu" activation function. The EEG model has 6 hidden layers with 256, 128, 64, 32, and 16 neurons, using the "Relu" activation function. The facial model has 4 hidden layers with 256, 128, 64, and 16 neurons, using the "Relu" activation function. Each of these three sub-models will output cognitive load assessment results based on their respective data inputs. Finally, these

three models will combine their outputs as inputs to a decision model. The decision model is an artificial neural network with 4 hidden layers having 128, 64, 32, and 16 neurons, respectively, and using the "Relu" activation function. The decision model will produce the final cognitive load assessment result based on the outputs from the three sub-models.

4. Experimental design

4.1. Environment

This study focuses on asynchronous online learning, which is one of the most widely used online learning format, specifically involving self-paced learning in an open online course. The learning behavior of participants takes place in a controlled laboratory environment. Participants sit at a desk, facing a Tobii T120 eye-tracking device, which not only collects eye-tracking data but also serves as an external monitor for presenting learning resources. This eye-tracking device is connected to a Lenovo laptop, and the laptop's built-in monitor display is observed and monitored by researchers. However, researchers do not intervene in the participants' actions during the learning process. Participants operate the system using external keyboards and mice. Above the eye-tracking device, there is a high-definition 2K resolution camera for video recording of participants' face. Participants also wear a brainwave headset based on the TGAM module to collect EEG signals. This EEG device has only one electrode and uses Bluetooth to transmit filtered EEG data to the computer, causing minimal interference with the learning process.

4.2. Materials and tasks

In countries or regions where English is not the native language, learning English is often a crucial aspect of education. The foundation of English learning lies in vocabulary acquisition, making it of utmost importance. Therefore, this study selected an online English vocabulary course as the experimental material, which serves as the learning material for participants. The course is divided into multiple sections, and in the original course sequence, the sections are arranged in increasing order of difficulty. For this research, several sections from the course were selected as experimental materials. These sections cover a range of difficulty levels, from easy to difficult, which may lead to participants having different cognitive loads. Participants, in the experimental environment, are required to independently complete the specified learning tasks on the computer.

Upon completing each section of the course material, participants are required to take a test that serves two primary objectives. Firstly, it aims to assess the effectiveness of their learning by evaluating their understanding of the section's content, including aspects such as word parts of speech, spelling, and meanings. Secondly, it measures the cognitive load experienced by learners during their study of the section's content. To evaluate cognitive load, a widely

Table 1. Features extracted from each modality of data.

Data modal		Feature extraction
Eye tracking	Full screen	Mean pupil diameter of the left eye (MPDL), Standard deviation of the pupil diameter of the left eye (StdPDL), mean pupil diameter of the right eye (MPDR), standard deviation of the pupil diameter of the right eye (StdPDR), Saccade distance (SD), the time of first fixation in the image area (TFF_Img), total fixation count (TFC), Total fixation duration (TFD), Fixation count in image areas (FC_Img), fixation duration in image areas (FD_Img), fixation count in text areas (FC_Text), fixation duration in text areas (FD_Text), proportion of fixation count in image areas to the total fixation count on the screen (PFC_Img/Sc), Proportion of fixation duration in image areas to the total fixation duration on the screen (PFD_Img/Sc), Proportion of fixation count in text areas to the total fixation count on the screen (PFC_Text/Sc), proportion of fixation duration in text areas to the total fixation duration on the screen (PFD_Text/Sc)
	AOIs	Fixation count in image areas (FC_Img), fixation duration in image areas (FD_Img), Fixation count in text areas (FC_Text), fixation duration in text areas (FD_Text), Proportion of fixation count in image areas to the total fixation count on the screen (PFC_Img/Sc), Proportion of fixation duration in image areas to the total fixation duration on the screen (PFD_Img/Sc), Proportion of fixation count in text areas to the total fixation count on the screen (PFC_Text/Sc), proportion of fixation duration in text areas to the total fixation duration on the screen (PFD_Text/Sc)
EEG		The mean, standard deviation, maximum, minimum, mean of the absolute first-order difference, and mean of the absolute second-order difference for Delta, Theta, LowAlpha, HighAlpha, LowBeta, HighBeta, LowGamma, MiddleGamma frequency bands, as well as attention and meditation
Face	Action unit	The mean and standard deviation of action unit (AU) intensity for AU01, AU02, AU04, AU05, AU06, AU07, AU09, AU10, AU12, AU14, AU15, AU17, AU20, AU23, AU25, AU26, AU45
	Emotion	The total number of occurrences for positive, neutral, and negative Expressions

recognized cognitive load scale known as the Paas Scale is employed. This scale consists of two questions designed to assess cognitive load from two dimensions: task difficulty and mental effort. The results of this scale are only used for the supervised training process and model performance testing, and are no longer needed in practical use.

4.3. Data collection

This study involved the collection of eye-tracking, EEG signals, and facial expression data from learners during the learning process. Based on data from these three modalities, feature extraction was conducted separately, as illustrated in Table 1.

The eye tracker, operating at a 60 Hz sampling rate, continuously recorded the learner's eye movement data throughout the session. To comprehensively analyze the distribution of the learner's visual attention, we divided the visual elements in the learning materials into two distinct regions: image areas and text areas. For both the entire screen and these specific regions, we calculated a wide range of eye movement metrics, including mean pupil diameter of the left eye (MPDL), standard deviation of the pupil diameter of the left eye (StdPDL), mean pupil diameter of the right eye (MPDR), standard deviation of the pupil diameter of the right eye (StdPDR), saccade distance (SD), the time of first fixation in the image area (TFF_Img), total fixation count (TFC), total fixation duration (TFD), fixation count in image areas (FC_Img), fixation duration in image areas (FD_Img), fixation count in text areas (FC_Text), fixation duration in text areas (FD_Text), proportion of fixation count in image areas to the total fixation count on the screen (PFC_Img/Sc), proportion of fixation duration in

image areas to the total fixation duration on the screen (PFD_Img/Sc), proportion of fixation count in text areas to the total fixation count on the screen (PFC_Text/Sc), and proportion of fixation duration in text areas to the total fixation duration on the screen (PFD_Text/Sc).

The EEG headset, based on the TGAM module, recorded the learner's EEG signals at 1-second intervals and processed the raw signals into different frequency bands using its built-in chip. These frequency bands include Delta, Theta, LowAlpha, HighAlpha, LowBeta, HighBeta, LowGamma, HighGamma, as well as Attention and Meditation computed by the embedded eSense algorithm. Attention and Meditation represent levels of focus and relaxation, each quantified within a range of 0 to 100. For feature extraction from the EEG signals, we computed statistical features for the five frequency bands and Attention and Meditation, including mean, standard deviation, maximum, minimum, mean of the absolute first-order difference, and mean of the absolute second-order difference. Due to the device's low sampling rate of 1 Hz, it is challenging to extract frequency domain information; therefore, only time-domain features were extracted from the EEG data.

During the learning process, a high-definition camera was used to continuously capture the learner's facial changes, and this data was saved as video files at a rate of 30 frames per second. The OpenFace tool was employed to detect the learner's facial action units from the recorded video. Additionally, each frame of the camera-recorded video was input into a facial expression recognition model, allowing for the identification of all facial expressions displayed by the learner during the learning process, which were categorized into three types: positive, neutral, and negative. The final set of extracted features includes the mean and standard deviation of the intensity for each facial

action unit and the total counts of the three categories of facial expressions.

Due to the different sampling frequencies of data from different modalities, the size of the generated dataset also varies. The eye tracker has a sampling frequency of 60 Hz, generating 60 data points per second, each containing the 24 features mentioned above. The EEG sampling rate is slow, only 1 Hz, producing 1 data per second, each containing 10 features. The Face sampling rate is 30 Hz, generating 30 data points per second, each containing 18 features.

Overall, 91 pieces of data are generated per second, and the length of each word learning is 15 seconds, which means that each experiment will generate 1365 pieces of data, corresponding to the cognitive load of one participant.

4.4. Participants

This study chose to convene participants within the scope of our university. The participants in this study had normal unaided or corrected vision, and they had recently undergone a university-mandated entrance medical examination, which confirmed that all their physical health indicators were normal and that they had no history of neurological disorders. They were in the age range of 21~25 years old ($M = 23.7$, $SD = 1.6$) and were all university students, totaling 220 individuals. Due to excessive visual correction or other potential causes weakening the effectiveness of eye tracking, we only selected participants with eye tracking effectiveness higher than 70% in the preliminary test to participate in the experiment. Finally, 194 individuals met the requirements. They all voluntarily signed a written consent form to participate in the experiment. In addition, the experiment was approved by the local institutions and university review board (Approval No. HR513-2021).

4.5. Procedure

1. Participants arrived at the experimental location at the scheduled time, where researchers informed them about the purpose, procedure, and instructions of the experiment, recorded their basic information, and obtained their informed consent.
2. Researchers assisted participants in wearing the necessary hardware devices and adjusted the software environment.
3. Participants independently operated the computer to complete the designated course content.
4. After completing each sub-course, participants took a test to assess their learning outcomes and self-reported their overall cognitive load using the Paas Cognitive Load Questionnaire.
5. The experiment concluded for each participant after completing the learning of five sub-courses and the corresponding post-tests.

5. Result

5.1. Model evaluation results

The experiment resulted in a total of 13.2 thousand facial expression data points, 26.4 thousand eye-tracking data points, and 4.4 thousand EEG data points. These extensive data from the three modalities underwent feature extraction and were combined to create a feature matrix for building the assessment model.

In this study, self-reported cognitive load values provided by learners were categorized into three levels: low, medium, and high, and all data were split into training and testing sets at a 7:3 ratio. The training set was used for model training, while the testing set was employed to assess the final model performance. Furthermore, to evaluate the impact of data modality selection and multimodal fusion methods on model performance, a comparison was made between multimodal decision-level fusion, multimodal feature-level fusion, and the evaluation accuracy on the testing set when using individual modalities, as shown in Figure 2. Firstly, individual modality accuracies were examined. Among the three single-modal models within the framework proposed in this study, the eye-tracking modality achieved the highest accuracy at 86.44%. In contrast, the facial and EEG modalities achieved lower accuracies at 67.79 and 61.01%, respectively, compared to the eye-tracking modality. In the case of multimodal models based on multiple data sources, both decision-level fusion and feature-level fusion outperformed the single-modal models. The decision-level fusion model achieved an accuracy of 91.52%, while the feature-level fusion model achieved an accuracy of 88.13%. It is evident that in the context of multimodal fusion, decision-level fusion surpasses feature-level fusion in terms of accuracy.

5.2. Correlation analysis

The feature extraction used in the modeling process is implemented by machine learning algorithms, and to explain the relationship between the feature dimensions of multimodal data and cognitive load, correlation coefficient analysis can be used.

Tables 2 and 3 present the correlations between various eye-tracking metrics and learners' cognitive load during the learning process. Firstly, in Table 2, for the analysis of pupil dilation characteristics, the metrics related to pupil diameter did not show significant or high correlations. While pupil dilation and constriction are physiologically related to cognitive activities and attention, they are highly susceptible to other factors such as lighting conditions and age (Duchowski et al., 2018; Winn et al., 1994). This study also has other findings, upon reviewing the participants' recordings, it was observed that some participants tended to become more distracted when faced with high cognitive load learning content. These factors may have contributed to the relatively low correlation between pupil-related metrics and cognitive load. Notably, the time of the first fixation in the image area exhibited a strong and significant correlation with cognitive load, with a correlation coefficient of 0.723,

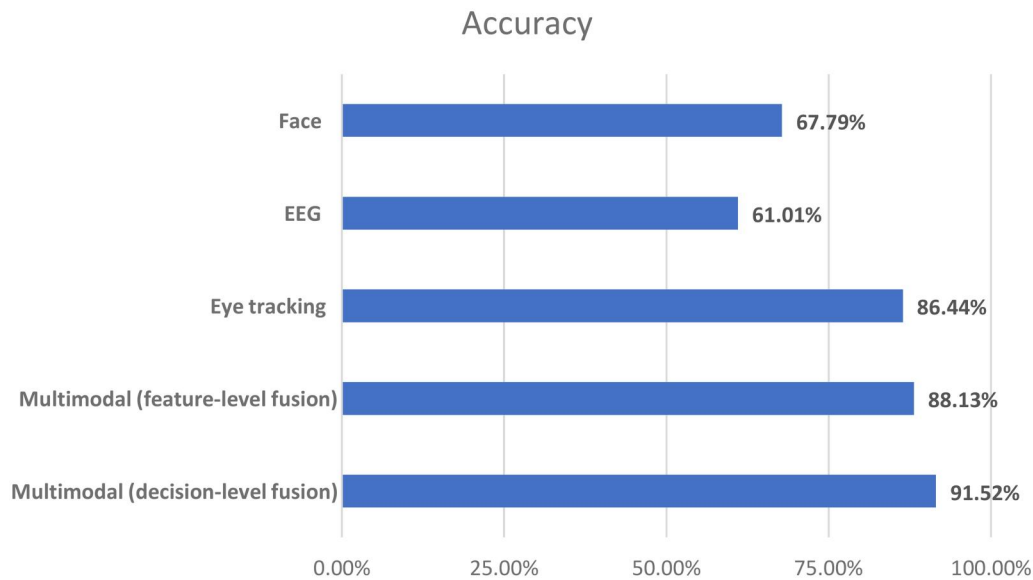


Figure 2. Comparison of model evaluation results.

Table 2. Correlations between eye-tracking metrics in full screen and cognitive load.

	StdPDL	StdPDR	MPDL	MPDR	TFF_img	SD	TFC	TFD
CL	0.023	0.013	0.080	0.115	0.723**	0.193**	0.274**	−0.042

** $p < 0.01$.

indicating that learners tend to notice the picture area later when confronted with more challenging content. Table 3 presents the correlations between cognitive load and eight eye-tracking metrics related to both picture and text areas, as well as their proportions relative to the full screen. All metrics showed strong and significant correlations, with absolute correlation coefficients exceeding 0.8. Comparing conventional statistical metrics to proportion-based metrics, the proportion-based metrics exhibited stronger correlations. This is because eye-tracking devices may lose or misinterpret a small amount of data due to factors like blinking or angles during operation. Consequently, despite all learners having the same learning duration, there may be slight variations in the effective length of eye-tracking data, which is an unavoidable situation. Therefore, for metrics within different areas of interest, calculating their relative values is more effective than calculating absolute values. Additionally, metrics in the picture area showed negative correlations, while those in the text area showed positive correlations. This suggests that when cognitive load is higher, learners tend to focus more on text, whereas when cognitive load is lower, they tend to pay more attention to images.

Table 4 displays the correlation between the average signals of various sub-frequency bands in the sub-EEG and cognitive load. From the strength of the correlation, it can be observed that the signals of the Delta, HighAlpha, and LowBeta frequency bands are negatively correlated with cognitive load and the correlations are significant. The respective correlation coefficients are -0.448 , -0.276 , and -0.386 , with Delta frequency band having the strongest correlation. On the other hand, the signal in the LowGamma frequency band is significantly positively correlated with cognitive

load, with a correlation coefficient of 0.232. Within EEG signals, slower-frequency bands are more closely associated with a person's subconscious activities, especially the Delta frequency band, which typically exhibits stronger signals during deep sleep. Additionally, previous research has suggested that Delta and Alpha frequency bands are linked to a person's fatigue level (Bon et al., 2012; Gharagozlou et al., 2015; Li et al., 2020), which is consistent with our study. On the other hand, higher-frequency Gamma signals are often associated with advanced cognitive activities. In our study, when learners faced challenging learning tasks, the signal strength in the LowGamma frequency band increased. This is because during such moments, individuals are in a state of active learning, and their brains are continuously receiving and processing information. Consequently, the signal strength in the corresponding lower-frequency bands decreased.

In this study, the OpenFace tool was used to detect the intensity of different facial actions. Table 5 presents the correlations between the average intensity of selected action units and cognitive load. Although the correlation between facial action and cognitive load is not high, we still conducted some analysis on this. Only a subset of action units is displayed because the facial action correlations yielded much lower values compared to other modalities, with most action units having absolute correlation coefficients less than 0.1. Therefore, only a few representative action units are shown. The action units with relatively higher correlations are AU_4, AU_10, and AU15, with correlation coefficients of 0.151, 0.271, and -0.162 , respectively. These action units correspond to eyebrow lowering, upper lip raising, and mouth corner lowering. Eyebrow lowering may indicate that learners tend to furrow their brows when encountering challenging learning materials. The other two mouth-related actions may be attributed to learners subvocalizing or silently repeating text content during the learning process. These actions are more likely to occur when the content is challenging, as suggested by the correlations with eye-

Table 3. Correlations between eye-tracking metrics in AOIs and cognitive load.

	Conventional				Proportional			
	FC_Img	FD_Img	FC_Text	FD_Text	PFC_Img/Sc	PFD_Img/Sc	PFC_Text/Sc	PFD_Text/Sc
CL	−0.731**	−0.759**	0.564**	0.481**	−0.811**	−0.813**	0.807**	0.808**

** $p < 0.01$.

Table 4. Correlation between the mean value of EEG signals in each frequency band and cognitive load.

	Delta	Theta	LowAlpha	HighAlpha	LowBeta	HighBeta	LowGamma	MiddleGamma
CL	−0.448**	−0.161*	0.005	−0.276**	−0.386**	0.141	0.232**	−0.085

* $p < 0.05$; ** $p < 0.01$.

Table 5. Correlation between certain facial movements and cognitive load.

	AU_4	AU_10	AU_15
CL	0.151*	0.271**	−0.162**

* $p < 0.05$; ** $p < 0.01$.

tracking indicators. In contrast, when the content is simpler, learners are less likely to engage in such subvocalization actions, as indicated by the lower correlations with eye-tracking measures.

6. Discussion

6.1. Calculating the proportional values of eye-tracking metrics in the AOIs is more advantageous

Eye-tracking technology is a method used to measure eye movements and gain insights into human cognition, emotions, and physiological states in various scenarios (Martinez-Marquez et al., 2021). This technique relies on cameras or sensors to track corneal light reflections and pupil positions. The cornea reflects infrared light, while the pupil absorbs it. By tracking the position and angles of these reflections, eye-tracking devices can determine the movement of the eyeballs (Bott et al., 2020). However, this operational principle inevitably leads to occasional data loss during recording due to factors such as excessive head movement, strong lighting, and blinking, making it difficult to record some data points. As a result, there may be slight variations in the effective data length collected from different learners experiencing the same learning content (Chan et al., 2022). Although this effect is minor, it has been commonly overlooked in a large body of eye-tracking research. This bias can indeed have some impact on the statistical metrics within the AOIs.

To solve the research question 1, we focus on how to effectively construct a cognitive load assessment model. The experience here tells us that using ratios is more effective than absolute values. Hence, in this study, when calculating the fixation count and fixation duration in the two AOIs, namely text and images, we additionally computed the proportion of these metrics within the full screen. The results in Table 3 show that proportional metrics exhibit significantly stronger correlations with cognitive load compared to traditional direct statistical metrics. This underscores that calculating relative values can mitigate, to a certain extent, the impact of technical issues related to the eye-tracking device on the data results.

6.2. Multimodal outperforms unimodal and decision-level fusion outperforms feature-level fusion

In recent years, with the advancement and widespread availability of various sensor technologies, these devices have become increasingly convenient and cost-effective. This development has made it possible to assess cognitive or learning states based on multimodal data. Numerous studies have shown that models constructed using multimodal data tend to outperform unimodal models in terms of performance (Herbig et al., 2021; Wilson et al., 2021). Our study aligns with this trend, as observed in the model evaluation results in Figure 2 for research question 1. Within the evaluation framework proposed in this study, all multimodal assessment models performed better than their respective unimodal counterparts. This is because data from different modalities can offer complementary information, thereby enhancing the accuracy and robustness of the models (Gammulle et al., 2021; Li et al., 2022). Moreover, our study found that decision-level fusion models outperformed feature-level fusion models. However, it's important to note that there is no absolute superiority or inferiority between these two fusion methods. The choice between them depends on the specific application scenarios and data characteristics.

6.3. Multimodal data analysis results contribute to optimizing online learning

Collecting multimodal data from online learners not only serves the purpose of assessing cognitive load but also provides valuable insights that empower educators to understand learners' states and enhance the optimization of online learning. Recording learners' eye-tracking, EEG signals, and facial expressions not only allows learners to gain self-awareness through dashboards but also assists educators in tracking learner states more objectively. This data helps in evaluating learner engagement, attention levels, and more (Al-Nafjan & Aldayel, 2022). Additionally, for research question 2, it enables educators to promptly identify students at risk of dropout or course failure through outlier data. Furthermore, our research revealed that when learners focus more on image areas, their cognitive load tends to be lower, while the opposite is true when they concentrate more on text areas. This suggests that, to prevent cognitive overload, instructional designers can complement challenging learning resources with explanatory images, effectively reducing learners' cognitive load. This finding aligns with conclusions

from prior research (Kaplan-Rakowski et al., 2021). In general, analyzing eye-tracking data from different areas also allows educators to validate the effectiveness of specific design elements in learning resources (Liu, Zhang et al., 2023). For example, they can determine whether certain visual elements meant to guide students' attention are effective and whether learners pay more attention to critical content. Based on these analytical insights, educators can make more precise optimizations to online learning resources.

7. Conclusion

In the current era of widespread online learning, this study is dedicated to the real-time assessment of learners' cognitive load within online learning environments. To achieve this goal, we propose a research framework for assessing cognitive load in online learning, which leverages three modalities of data: eye-tracking, EEG, and facial expressions. Based on this framework, we conducted practical data collection in online learning environments and developed an assessment model. First and foremost, this study provides a reference guideline for future research by introducing this research framework. Furthermore, the model constructed in this study has been validated to achieve an accuracy rate of 91.52%. As various sensor devices become more prevalent, cost-effective, and convenient, this model can be applied to intelligent online learning systems or platforms, enabling real-time assessment of learners' cognitive load (research question 1). Additionally, the results of multimodal data analysis will guide the development of features and resources within online learning platforms, as well as the formulation of strategies for regulating cognitive load (research question 2). Ultimately, this will lead to adaptive online learning based on cognitive load.

In the trial of the model, the user pointed out that the model also needs a feedback system, so that the user can timely know their cognitive load situation. Therefore, there are some limitations to this study, such as limited application topics and lack of feedback systems. Future research will focus on model fine-tuning, complex learning scenarios and themes designing and expansion of research scale to enhance the model's generalization capabilities.

Acknowledgements

To the peers and reviewers who have provided suggestions on this paper!

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Natural Science Foundation of Shanghai, China under [Grant 22ZR1421300].

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